Geographic coverage debt of frozen Earth-observation foundation models: an encoder-robustness reliability diagnostic and a conditional target-recalibration remedy

Zeyu Fu

***Abstract*—Frozen geospatial foundation models (GFMs) are increasingly deployed as off-the-shelf encoders for land-cover and local-climate-zone (LCZ) monitoring, yet practitioners lack guidance on which frozen encoder to trust after a regional move, or on how to restore a reliability guarantee once shift breaks it. We supply both through a conformal reliability audit of frozen GFMs under real cross-region shift (thirteen encoders on GEO-Bench BigEarthNet-S2 for the FNR analysis; five on EuroSAT-MS (1) and non-European m-so2sat), yielding findings that shifted accuracy and calibration do not surface. First, under a multi-label false-negative-rate (FNR) risk-control guarantee that every frozen encoder meets in-distribution, all thirteen encoders incur a real geographic debt under shift (FNR 2.6–$4.7\times$ nominal), repaired only by a labeled target-region slice; and accuracy poorly predicts which encoder is most robust—the least accurate is the most FNR-robust (0.130), but at n=13 the rank correlation is a weak, non-significant positive ($\mathrm{Spearman}(\mathrm{FNR_{sh}}, \mathrm{mAP}) = +0.32$, p=0.28), not the strong $+0.90$ inversion a five-encoder slice suggested. Second, as a conditional finding, restoring the *marginal* coverage guarantee can hide a per-class collapse—one land-cover class is covered at 0.850 under a 0.956 marginal (the same class every seed; range 0.826–0.871; n=5). Target-side recalibration on a small labeled slice restores both guarantees (e.g., Clay $0.176 \rightarrow 0.022$ at α=0.05) and ranks the encoders by coverage debt. We also map the audit's validity boundary: single-label prediction sets collapse to singletons above accuracy ≈ 0.80. The audit is diagnostic and reuses existing conformal estimators; the preregistration outcome and exploratory-analysis disclosures are in §V.**

***Index Terms*—Earth observation, geospatial foundation models, conformal prediction, distribution shift, uncertainty quantification, reliability, calibration.**

I. Introduction

GEOSPATIAL foundation models are pretrained once and reused frozen as feature extractors for many downstream tasks. Beyond the encoders audited here (thirteen on BigEarthNet, five on EuroSAT and m-so2sat), frozen GFM backbones now span single-sensor optical pretraining (SatMAE (2), Scale-MAE (3), SeCo (4), GFM (5)), multi-sensor and multi-modal pretraining (SkySense (6), AnySat (7), TerraFM (8), Galileo (9)), and pixel-timeseries pretraining (Presto (10)); none of these families is conformally audited under geographic shift, and our roster (Methods) is illustrative of the diagnostic rather than a leaderboard survey of this rapidly growing space. Their accuracy under geographic shift has been studied (11, 12). Calibration itself is not entirely unstudied: SHRUG-FM (11) evaluates SSL4EO ViT-S/16 (MoCo/MAE/DINO) checkpoints on burn-scar segmentation (ExEBench) and reports expected calibration error (ECE 0.0248–0.0328), but it has no conformal-prediction arm, no spatial-holdout cross-validation, and does not evaluate Clay or Prithvi. Uncertainty quantification for deep networks more broadly spans Bayesian, ensemble, and distribution-free approaches (13); conformal prediction is the distribution-free branch we build on, and it itself has entered Earth observation: Singh et al. (14) bring split and (class-conditional) Mondrian conformal prediction to EO pipelines (Google Earth Engine), GeoConformal (15) adds geographically-weighted conformal intervals for *spatial regression*, and Mao et al. (16) develop model-free conformal prediction for spatially indexed data under local exchangeability; none of the three audits frozen foundation-model encoders, nor coverage loss under an explicit geographic train/deploy shift. Weighted conformal methods for covariate shift (17) are the natural theoretical remedy but require density-ratio estimates we do not assume here. Closest in spirit, Fillioux et al. (18) ask whether frozen vision foundation models are good conformal predictors, but on in-distribution image classification with no geographic-shift audit; and Aolaritei et al. (19) develop Lévy–Prokhorov theory for conformal robustness to distribution shift—complementary theoretical machinery that they do not instantiate on Earth observation. Our geographic-shift framing is also distinct from the broader unsupervised-domain-adaptation literature for remote sensing classification (20, 21), which typically retrains or adapts the feature extractor itself against the target domain; we instead ask whether a *frozen*, never-retrained encoder's conformal guarantee survives a regional shift, and how to repair it with only a labeled target-region calibration slice, not further training. Consequently, *conformal coverage* reliability under shift—whether a conformal predictor's guarantee (22, 23) on frozen classification GFMs such as Clay and Prithvi survives a move to a new region—has received far less attention. We ask a sharp, falsifiable question: under a real geographic shift, does a conformal audit of a frozen GFM tell a practitioner anything that shifted accuracy and calibration do not? We find that it does, in two regimes, and we characterize precisely where it does not. Our primary contribution is a within-encoder reliability diagnostic that needs no cross-encoder statistics: under a conformal-risk-control (24) bound on the

Affiliation to be added. Correspondence e-mail: fuzeyu09@gmail.com.
Manuscript dates and funding acknowledgement to be added at submission.

example-averaged false-negative rate (FNR) on GEO-Bench BigEarthNet-S2, *every one* of thirteen frozen encoders meets the FNR guarantee in-distribution yet incurs a real geographic debt under shift (0.130–0.234, 2.6–4.7× nominal), which a labeled target-region slice repairs (0.002–0.036). On top of this, accuracy is a weak and unreliable guide to *which* encoder is most robust: the least accurate encoder (Prithvi, mAP 0.432) is the most FNR-robust (0.130) and the most accurate is not, but the rank correlation over the full roster is only a weak, non-significant positive (Spearman $\rho = +0.32$; Table I)—the strong $+0.90$ inversion visible in our first five encoders regresses toward zero when the roster is expanded, so we report it as directional evidence, not a proven inversion (§III). A companion FNR-versus-set-size tradeoff ($\rho = -0.66$ at α=0.10; same table) means robustness is bought with larger prediction sets, an operating-point choice no accuracy scalar can name. As a second, conditional finding, marginal coverage repair can hide a per-class collapse: on EuroSAT, after target-side recalibration restores the *marginal* guarantee, one land-cover class is still covered at only 0.850 under a 0.956 marginal—a failure invisible to accuracy and to ECE, reported as a single load-bearing case (Table II). Our third contribution is the singleton-degeneracy boundary, mapping where the audit is informative versus accuracy-restated: the marginal single-label set collapses to a singleton above accuracy ≈ 0.80, so EuroSAT at full data is an explicit *negative control* bounding the audit's validity regime (So2Sat/BigEarthNet sit below it). Our fourth, secondary and descriptive contribution is an encoder coverage-debt ranking and a target-recalibration remedy, with a negative-control battery that rules out the confounds a coverage-debt result could otherwise be dismissed as: the magnitude of geographic coverage debt ranks the frozen GFMs (Prithvi > DOFA (25) > SSL4EO-DINO > SSL4EO-MAE > Clay)—though, because the source-split predictor is in the singleton regime, this ordering coincides with the shifted-accuracy ordering and is reported as descriptive, not as evidence the conformal layer beats accuracy; covariate-shift-weighted conformal under-covers for every encoder, oracle and deployable label-shift reweighting fail (worsening coverage for three of four encoders, ruling out a pure class-prior effect and consistent with a within-class conditional shift), and spatial-Mondrian conditioning adds nothing measurable over non-spatial target calibration, so the operative axis is access to a small labeled target-region slice (the audit is diagnostic; scope and estimator provenance are stated in §V). We establish these on a real EuroSAT-MS (1) cross-region split with five frozen GFMs (the original four plus DOFA (25), added on the EuroSAT grid), corroborate them on GEO-Bench BigEarthNet-S2 (26) (a documented dominant-label proxy, a rigorous per-label multi-label conformal analysis, and a conformal-risk-control arm), and—to test generalization beyond Europe—on the non-European, multi-continent GEO-Bench m-so2sat benchmark. Our analysis is preregistered; where the preregistered *universal*-restoration criterion was not met we preserve that outcome verbatim (§V) and report the conditional, encoder-dependent result as its honest reframe. Fig. 1 summarizes the audit pipeline end to end.

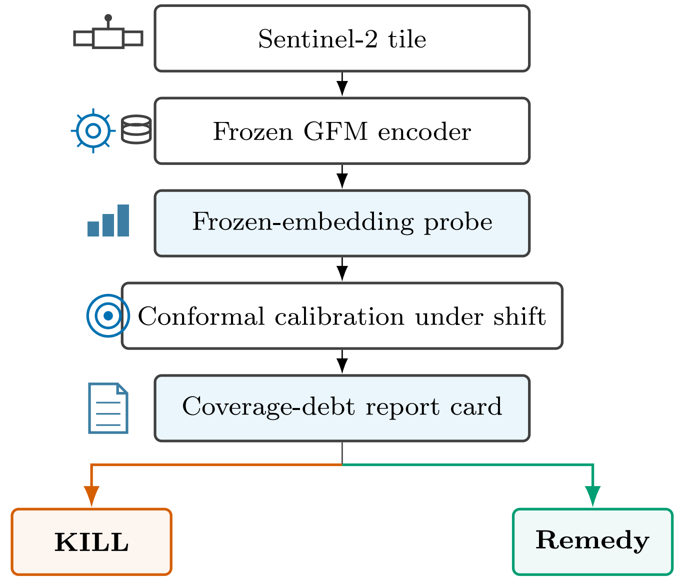

Fig. 1. Overview of the coverage-debt audit. A Sentinel-2 tile (EuroSAT-MS / GEO-Bench BigEarthNet-S2 / m-so2sat) is passed through one of five frozen, interchangeable GFM encoders (Prithvi, Clay, SSL4EO-DINO, SSL4EO-MAE, DOFA) and a frozen-embedding probe (logistic for the single-label tasks, multi-label otherwise), then conformally calibrated under the real geographic (region-transfer) shift—source split-conformal vs. spatial-Mondrian target-region calibration, source = E/Central Europe, target = W/Iberian-Atlantic Europe—yielding per-class conformal sets, producing a coverage-debt report card (marginal vs. per-class conditional coverage; FNR under shift). The preregistered universal-restoration criterion returns KILL; the validated remedy is a conditional, encoder-dependent target-region recalibration (§V).

II. Methods

The data come from EuroSAT-MS (1) (Sentinel-2 L2A, 13 bands, 27,000 tiles, 10 classes). The shift is a *real* geographic cross-region holdout using per-tile longitude: source = E/Central Europe (lon $\geq P_{33}$), target = W/Iberian-Atlantic Europe (lon $< P_{33}$); no image transform is applied. The P_{33} cut is a single, fixed longitude-percentile boundary chosen once in advance; a post-hoc boundary-sensitivity sweep over $P_{25}/P_{40}/P_{50}$ is reported in Results (Table VI; see Limitations). This cut also induces a class-frequency (label-prior) shift between source and target that we did not adjust for in the headline logistic probe: at P_{33}, class 5 goes from 2.62% of source tiles to 17.13% of target tiles (6.54× enriched), class 1 from 15.02% to 3.16% ($\sim 4.7\times$ depleted), class 8 from 11.60% to 4.51%, and class 9 from 14.10% to 5.04% (remaining classes shift by <2.4×; Table X). We re-fit the probe with class-balanced sample weighting on the same cached embeddings as a control (Table XI; see Discussion). The frozen encoders are Prithvi-EO-2.0-300M (27) (mean-pooled patch tokens, 1024-d); Clay v1.5 (28) (cls token, 1024-d); SSL4EO-S12 (29) ViT-S/16 DINO via torchgeo (cls token, 384-d); SSL4EO-S12 (29) ViT-B/16 MAE ep99 (`wangyi111/SSL4EO-S12` on HuggingFace, 87.8M params, token-free global pool, 768-d). A fifth encoder, DOFA (Dynamic One-For-All) (25)—a ViT-B/16 whose dynamic wavelength-conditioned patch embedding ingests the Sentinel-2 bands natively (`dofa_base_patch16_224`, `DOFA_MAE` weights via `torchgeo`)—was added on the EuroSAT grid only, frozen and probed identically. The original four weights loaded strictly (zero missing keys), verified real (in-distribution probe accuracy 0.95–0.99 vs. chance 0.10). For the BigEarthNet FNR analysis—the one cross-encoder comparison that turns on statistical power—we expand this battery to *thirteen* frozen encoders by adding eight more loaded through first-party trusted paths (`torchgeo` 0.9.0 built-in weight builders, no third-party repository code executed), each with normalization read from the weight's own shipped transform: SSL4EO-S12 MoCo (ViT-S/16, 384-d) and its ResNet50 MoCo/DINO variants (2048-d) (29); SoftCon (30) ViT-B/14 (768-d) and ViT-S/14 (384-d); an SSL4EO MAE ViT-L/16 (1024-d); and CROMA (31) Base (768-d) and Large (1024-d). Two further planned arms (`dofa_large`, `prithvi_600m`) were dropped by the pre-specified smoke/mAP gates (missing or non-finite feature caches at extraction), leaving thirteen scored; the three anchor encoders reused the frozen caches and reproduce the earlier five-encoder records exactly. The expanded roster is used only for the BigEarthNet CRC comparison; EuroSAT and m-so2sat retain the five-encoder battery. A logistic probe is trained on frozen embeddings. We compare source-fit split-conformal (22, 23) (LAC (32) nonconformity $1-\hat{p}_{\text{true}}$) against a *spatial-Mondrian* conformal predictor (33) that draws per-spatial-bin quantiles from a disjoint same-region target-calibration slice. We adopt the LAC score for its simplicity; adaptive prediction sets (34) are a natural alternative nonconformity score we do not explore. The Mondrian and target-calibration arms rely on exchangeability within each spatial/target bin, a weaker requirement than global exchangeability; more general non-exchangeable-shift theory (35) formalizes how conformal coverage can degrade under shifts broader than the fixed geographic partition studied here, and the conformal-risk-control arm we use below (24) is itself an instance of the general distribution-free risk-controlling framework of Bates et al. (36). We sweep $\alpha \in \{0.05, 0.10, 0.20\}$, 5 seeds, and bootstrap the paired per-seed restoration statistic $|\text{cov}_{\text{split}} - \text{nom}| - |\text{cov}_{\text{Mondrian}} - \text{nom}|$ (CI $> 0 \Rightarrow$ restored). A post-hoc boundary-sensitivity sweep repeats the full protocol at $P_{25}/P_{33}/P_{40}/P_{50}$ longitude cuts on the same cached embeddings, and adds a third, *label-access ablation* arm: non-spatial split conformal whose single LAC quantile is drawn from the same labeled target-region calibration slice the Mondrian arm uses (isolating target-label access from spatial conditioning). All figures (Fig. 3–5, 6, 7) are generated directly from the frozen result records by the accompanying figure-generation scripts. As a post-hoc, exploratory extension, after the preregistered gate (§V) we added three decoupling analyses on the same frozen-feature caches, computed from cached embeddings with no GPU re-extraction: (i) on BigEarthNet-S2, a per-encoder comparison of the CRC arm's shifted FNR against multi-label accuracy (macro mAP) and calibration (mean per-label ECE) analogs on the same footing (the recomputed shifted FNR reproduces the frozen CRC JSON, e.g. Prithvi 0.129 vs. 0.130); (ii) on EuroSAT, per-class conditional coverage of the spatial-Mondrian-repaired target sets at matched marginal coverage ($\approx$ 0.95–0.96), summarized by the worst-class gap $\big((1-\alpha)-\min_k \text{cov}_k\big)$; and (iii) a low-shot sweep that subsamples the EuroSAT probe-training fraction to $\{0.01, \dots, 1.0\}$ and records split set size and the coverage$-$accuracy gap, mapping the singleton-degeneracy boundary. Correlations across encoders are Spearman rank statistics (n=13 FNR on the expanded BigEarthNet roster, n=5 conditional coverage) and are reported as directional evidence, not proofs of independence (§V).

III. Results

A conformal reliability audit is worth its complexity only if it tells a practitioner something accuracy and calibration do not, so we lead with the regimes in which it does. Primary result (next): under a multi-label false-negative-rate (FNR) risk-control guarantee, all thirteen frozen encoders incur a real geographic FNR debt that only a labeled target slice repairs, and accuracy is a weak, unreliable guide to which encoder is most robust. As a conditional finding: on EuroSAT, restoring the *marginal* coverage guarantee hides a per-class conditional-coverage collapse that neither accuracy nor ECE predicts. We then characterize the audit's validity boundary: in the marginal single-label singleton regime the set-size axis is a deterministic restatement of accuracy, so EuroSAT at full data is a mapped negative control, not a decoupling. Only then do we report the encoder coverage-debt ranking and its target-side recalibration repair, as a secondary, descriptive diagnostic. Every axis-decoupling analysis below is post-hoc and exploratory (added after the preregistered gate; §V).

A. Primary result: a marginal FNR guarantee incurs a geographic debt no encoder escapes, and accuracy does not reliably predict which encoder is most robust

On GEO-Bench `m-bigearthnet` (BigEarthNet-S2; 7,669 tiles, 39 active labels; real E$\rightarrow$W geographic shift), a conformal-risk-control (CRC) predictor (24) bounds the example-averaged FNR—a set-valued guarantee with *no top-1-accuracy analog by construction*. We audit *thirteen* frozen encoders here (the five-encoder battery plus eight added at pull expressly to power the cross-encoder comparison; roster in §II). The within-encoder finding is unambiguous and requires no cross-encoder statistics: for *every one* of the thirteen encoders, source-calibrated CRC meets the target FNR in-distribution (0.051–0.056 at α=0.05) yet incurs a real FNR debt under shift (0.130 Prithvi to 0.234 ResNet50-MoCo, i.e. 2.6–4.7$\times$ nominal across the whole roster), which a target-side Mondrian-CRC arm repays (0.002–0.036; Table I, full per-α CRC/repair numbers in Table VII). Unlike the inert EuroSAT-singleton regime (validity-boundary result below), the repair is active here. This debt-and-repair is the reliability-relevant result: the marginal guarantee a practitioner would trust degrades under a regional move—uniformly, across every backbone we tested—and is restored only with a labeled target-region slice. This pattern is not merely descriptive: treating the thirteen preregistered per-encoder point estimates as a population, a distribution-free sign test confirms that shifted CRC FNR exceeds the nominal target in all thirteen encoders (two-sided $p \approx 2.4 \times 10^{-4}$) and that target-side Mondrian-CRC returns

FNR to at or below nominal in all thirteen (same test, same p); a Wilcoxon signed-rank sensitivity check agrees exactly (one-sided $p \approx 1.2 \times 10^{-4}$ for both directions, the smallest value attainable at n=13), and every one of the thirteen per-encoder bootstrap confidence intervals excludes nominal in the claimed direction, for both the debt and the repair arm. We report the one-sided figure only as a sensitivity check: the debt-and-repair direction was not discovered in this roster run but inherited from the original five-encoder confirmatory arm, so it is defensible, but the two-sided value is what we use as the headline since it needs no such defense and is already decisive. This is a population-level summary computed post-hoc over the frozen, preregistered per-encoder estimates—the per-encoder numbers are prereg and frozen, but the population test itself is not—and it supplies a second firm, statistically-tested positive result alongside the preregistered falsification of universal restoration (§V).

Does accuracy tell a practitioner *which* encoder to trust? Across the thirteen encoders the answer is no, though more weakly than our initial five-encoder probe suggested. The FNR under shift is only weakly rank-correlated with in-distribution mAP ($\mathrm{Spearman}(\mathrm{FNR}_{\mathrm{sh}}, \mathrm{mAP}) = +0.32$, permutation p=0.28, n=13, α=0.05; Fig. 2) and not at all with calibration ($\mathrm{Spearman}(\mathrm{FNR}_{\mathrm{sh}}, \mathrm{ECE}_{\mathrm{in}}) = -0.16$, p=0.60). The correlation is *positive*—higher accuracy never buys better FNR robustness, and the ordering never reverses to "better encoder $\rightarrow$ better reliability"—but it does not reach significance and 48 of 78 encoder pairs (62%) invert, so we do *not* claim a strong accuracy-to-reliability inversion. Two facts survive the expansion and carry the practical point. First, the extreme is intact: Prithvi, the least accurate encoder (mAP 0.432), is the *most* FNR-robust (0.130), while the two most FNR-fragile encoders—the ResNet50 CNNs (MoCo/DINO, 0.234/0.232)—sit mid-pack on accuracy; and Prithvi vs. Clay is a near-calibration-matched dissociation (ECE 0.082 vs. 0.078, differing by 0.004, yet FNR 0.130 vs. 0.176, the worse-calibrated encoder holding the better FNR). Second, the risk-versus-efficiency tradeoff persists: FNR robustness is bought with larger prediction sets ($\mathrm{Spearman}(\mathrm{FNR}_{\mathrm{sh}}, \text{set size}) = -0.33$ at α=0.05, -0.66 (p=0.017) at α=0.10; Prithvi buys the best FNR with the largest sets, 15.5 labels/tile). The honest reading is that accuracy is *not* a reliable guide to conformal FNR robustness under shift—the most accurate encoder is not the most robust and the least accurate is among the best—but at n=13 this is a weak, directional statement, not the strong rank inversion the five-encoder slice appeared to show. In those first five encoders the same statistic read $+0.90$ (asymptotic p=0.037; its *exact* permutation p was already 0.083); the roster expansion regresses it toward zero, precisely the power test the expansion was run to perform, and we report the weakened result rather than the headline it displaced (§V).

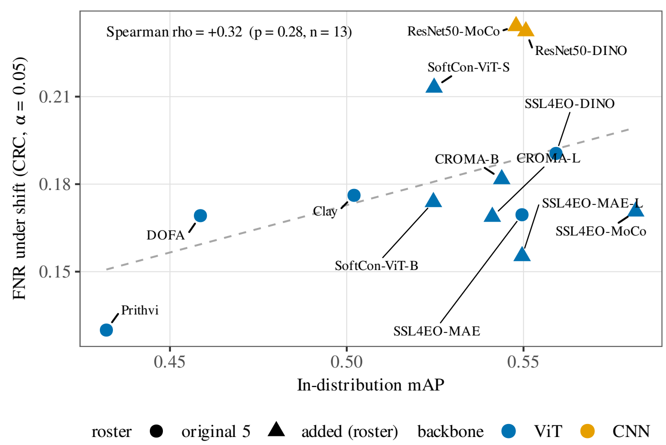

Fig. 2. The five-encoder inversion does not survive expansion to thirteen. FNR under shift vs. in-distribution mAP on `m-bigearthnet` (α=0.05, CRC arm); circles are the original five encoders, triangles the eight added at pull. The rank correlation is a weak, non-significant positive (Spearman $= +0.32$, permutation p=0.28, n=13; dashed line is an OLS guide only): accuracy never buys better FNR robustness, but the strong $+0.90$ inversion of the five-encoder slice regresses toward zero. Prithvi (least accurate) remains the most FNR-robust; the two ResNet50 CNNs are the most fragile at mid accuracy.

B. Secondary (conditional) result: marginal coverage repair hides a per-class conditional collapse

After the target-side spatial-Mondrian arm restores the *marginal* EuroSAT guarantee to $\approx$ 0.95–0.96 for every encoder (secondary results below), we ask what that marginal scalar hides: how badly does the worst individual land-cover class still under-cover? At matched marginal coverage (α=0.05, 5 seeds; Table II) the worst-class gap is scrambled relative to accuracy and ECE—$\mathrm{Spearman}(\text{worst gap}, \text{acc}) = +0.30$ (p=0.62), $\mathrm{Spearman}(\text{worst gap}, \mathrm{ECE}) = -0.30$ (p=0.62), 6 of 10 pairwise inversions against each, partial correlation $\approx$ 0 (-0.03 for acc$|$ece). The smoking gun: SSL4EO-DINO is mid-pack on accuracy (0.880, 3rd of 5) and ECE (0.076, 3rd) yet silently abandons a class—one EuroSAT class is covered at only 0.850 under a 0.956 marginal (a 0.106 within-encoder conditional shortfall, 2$\times$ the next-worst encoder's gap), and it is the same class that under-covers in all five seeds (per-seed coverage 0.826–0.871, std 0.016; the datum is seed-stable, not a single-seed artifact). Nothing in its accuracy or calibration predicts this: DOFA, the second-least accurate encoder (0.851), has the best conditional coverage (gap 0.031), while Clay—the most accurate (0.918) and best-calibrated (ECE 0.057)—has the second-worst (gap 0.048). The effect sharpens at α=0.10, where SSL4EO-DINO's worst class sits at 0.776 under a 0.926 marginal. Which frozen encoder quietly drops a class under geographic shift is invisible to accuracy and to ECE and is surfaced only by the conditional conformal audit. Honest caveat: n=5 gives low power—all $p > 0.28$ and the weak rank correlation even flips sign across α—so the claim is "no detectable/stable dependence plus large concrete inversions," not proven independence; the SSL4EO-DINO case is the load-bearing single datum and is reported as such (§V).

C. The audit's validity boundary: singleton degeneracy of the marginal single-label set

A conformal set-size axis can decouple from accuracy only where the sets are non-trivial. On EuroSAT at full data the source-calibrated split predictor emits *singletons* on

TABLE I

PRIMARY DECOUPLING ON GEO-BENCH M-BIGEARTHNET (α=0.05; 13 FROZEN ENCODERS, SORTED BY MAP): EVERY ENCODER MEETS THE TARGET FNR IN-DISTRIBUTION AND INCURS A 2.6–4.7× FNR DEBT UNDER SHIFT THAT THE TARGET-SIDE MONDRIAN-CRC ARM REPAIRS. "FNR SHIFT" IS THE SOURCE-CALIBRATED CRC ARM'S SHIFTED FNR; "REPAIRED" IS THE TARGET-SIDE MONDRIAN-CRC ARM; SET SIZE IS CRC LABELS/TILE UNDER SHIFT. ACCURACY IS A WEAK, NON-SIGNIFICANT GUIDE TO FNR ROBUSTNESS: $\mathrm{Spearman}(\mathrm{FNR_{sh}}, \mathrm{mAP}) = +0.32$ (p=0.28; 48/78 INVERSIONS), $(\mathrm{FNR_{sh}}, \mathrm{ECE}) = -0.16$ (p=0.60), $(\mathrm{FNR_{sh}}, \text{SET SIZE}) = -0.33$ (n=13; -0.66, p=0.017 AT α=0.10).

Encoder	mAP (in)	ECE (in)	FNR shift	FNR repaired	Set size
Prithvi	0.432	0.082	0.130	0.031	15.5
DOFA	0.459	0.057	0.169	0.021	13.0
Clay	0.502	0.078	0.176	0.022	13.3
SoftCon-ViT-B	0.525	0.033	0.174	0.031	11.9
SoftCon-ViT-S	0.525	0.020	0.213	0.036	10.7
CROMA-L	0.541	0.037	0.169	0.024	11.6
CROMA-B	0.544	0.046	0.182	0.030	12.1
ResNet50-MoCo	0.548	0.073	0.234	0.002	12.5
SSL4EO-MAE	0.550	0.037	0.170	0.030	11.8
SSL4EO-MAE (ViT-L)	0.550	0.038	0.155	0.029	11.9
ResNet50-DINO	0.551	0.055	0.232	0.014	10.8
SSL4EO-DINO	0.559	0.028	0.191	0.032	10.6
SSL4EO-MoCo	0.582	0.032	0.171	0.028	10.7

TABLE II

PER-CLASS CONDITIONAL COVERAGE OF THE MARGINALLY REPAIRED EUROSAT PREDICTOR (SPATIAL-MONDRIAN, α=0.05; MARGINAL COVERAGE MATCHED TO $\approx$ 0.95–0.96; 5 SEEDS). THE WORST-CLASS GAP DOES NOT TRACK ACCURACY OR ECE (Spearman = $+0.30/-0.30$, p=0.62; 6/10 INVERSIONS). SSL4EO-DINO, MID-PACK ON BOTH SCALARS, ABANDONS A CLASS AT 0.850 (SEED-STABLE; RANGE 0.826–0.871); THE MOST ACCURATE AND BEST-CALIBRATED ENCODER (CLAY) HAS THE SECOND-WORST GAP. ACCURACY/ECE COLUMNS ARE THE CANONICAL SHIFT VALUES; THE CONDITIONAL COLUMNS ARE THE EXP0 RECOMPUTE.

Encoder	Acc. shift	ECE shift	Marg. cov	Worst-class cov	Worst gap (vs. nominal)
DOFA	0.851	0.093	0.951	0.919	0.031
SSL4EO-MAE	0.898	0.065	0.962	0.912	0.038
Prithvi	0.833	0.109	0.955	0.905	0.045
Clay	0.918	0.057	0.960	0.902	0.048
SSL4EO-DINO	0.880	0.076	0.956	0.850	0.100

the shifted target (mean set size 1.00 for every encoder and α; per-seed ≤ 1.001), so its coverage coincides with shifted top-1 accuracy (5-seed-mean $|\Delta| \leq 2 \times 10^{-4}$) and the marginal set-size/efficiency axis carries no information beyond accuracy. Even after target-side repair the restored Mondrian set size is a deterministic image of accuracy loss: across the five encoders $\mathrm{Spearman}(\text{set size}, \text{acc}) = -1.0$ and $\mathrm{Spearman}(\text{set size}, \text{ECE}) = +1.0$ (α=0.05, zero pair-wise inversions). We therefore report set size only as a companion to a held guarantee, never as evidence that the conformal layer beats accuracy—the preregistered "efficiency-decoupling" route is empirically falsified on this benchmark and we say so. Mapping the boundary. Subsampling the probe-training fraction drives shifted accuracy down and locates where the audit switches from degenerate to informative (Table III). In the high-accuracy regime (acc $\gtrsim 0.80$ at α=0.05) sets are exact singletons (size $\equiv$ 1.000) and coverage $\equiv$ accuracy; as the task hardens below acc ≈ 0.78 the sets go non-singleton (size > 1.05) and the coverage$-$accuracy gap opens from 0 to $+0.22$. The more accurate the encoder, the higher the accuracy at which it degenerates (Clay singleton by acc $\approx$ 0.87, SSL4EO-DINO by $\approx$ 0.82). This makes EuroSAT at full data an explicit negative control: it sits above the boundary, so its *marginal* single-label audit is accuracy-restated—which is precisely why the per-class conditional audit above (on the non-singleton repaired sets) and the multi-label FNR audit on BigEarthNet are where the conformal layer earns its keep, and why So2Sat (17-class LCZ, shifted accuracy 0.52–0.60), which sits below the boundary, carries informative non-singleton sets even marginally (Table VIII). The boundary is itself a contribution: it tells a practitioner when a marginal conformal audit of a frozen GFM is worth running.

D. Secondary descriptive: the coverage-debt encoder ranking and its target-side repair

The remaining EuroSAT results are the coverage-debt ranking and target-recalibration repair. They are a useful descriptive diagnostic but, per the validity-boundary result above, the marginal ranking coincides with the shifted-accuracy ranking; we report them as such, not as an independent conformal signal. Coverage debt is real and encoder-dependent: under shift, accuracy drops $0.95 \rightarrow 0.83$ (Prithvi), $0.97 \rightarrow 0.88$ (SSL4EO-DINO), $0.98 \rightarrow 0.92$ (Clay), and ECE rises to 0.109 (Prithvi), 0.076 (SSL4EO-DINO), 0.065 (SSL4EO-MAE), 0.057 (Clay)—an ordering of calibration degradation across the four encoders that is preserved under the boundary sweep (Table VI) and the class-prior-balanced probe control (Discussion), so it does not rest on a bare four-point monotonicity (Fig. 3). As a continuous, bucket-free debt

TABLE III
SINGLETON-DEGENERACY BOUNDARY (EUROSAT, PRITHVI, α=0.05, 3 SEEDS): SUBSAMPLING THE PROBE-TRAINING FRACTION LOWERS SHIFTED ACCURACY AND MOVES THE SPLIT PREDICTOR FROM DEGENERATE (SINGLETONS, COVERAGE $\equiv$ ACCURACY) TO INFORMATIVE (NON-SINGLETON SETS, COVERAGE DECOUPLED FROM ACCURACY). THE TRANSITION SITS NEAR ACCURACY $\approx$ 0.78–0.80.

Train frac	Train n	Shifted acc	Split set size	Coverage $-$ accuracy
1.00	10854	0.832	1.000	$+0.000$
0.50	5427	0.805	1.026	$+0.011$
0.25	2713	0.783	1.072	$+0.026$
0.10	1085	0.765	1.202	$+0.066$
0.05	542	0.721	1.383	$+0.107$
0.02	217	0.645	1.866	$+0.196$
0.01	108	0.650	2.294	$+0.221$

measure we integrate the target coverage gap $\max(0, (1-\alpha) - \mathrm{cov}_{\mathrm{split}})$ over the α sweep: $D = 8.0/3.3/1.4/0.8 \times 10^{-3}$ for Prithvi / SSL4EO-DINO / SSL4EO-MAE / Clay—and $D \approx 6.2\times10^{-3}$ for DOFA, added on the same grid as a fifth encoder and the second-heaviest debt, between Prithvi and SSL4EO-DINO—recovering the same ordering without the coarse $k/3$ repaired-levels bucketing. The gap closes at $\alpha = 0.20$ for all five encoders because shifted accuracy already meets the 0.80 nominal—a no-debt artifact of the easy risk level, not evidence of a repair. DOFA's shifted split coverage is 0.851 at every risk level, so like the four battery encoders it carries under-coverage debt only at $\alpha = 0.05$ and 0.10.

Spatial-Mondrian repays the debt where it exists: at $\alpha = 0.05$ (nominal 0.95), source split-conformal under-covers (0.833 Prithvi, 0.851 DOFA, 0.880 SSL4EO-DINO, 0.898 SSL4EO-MAE,[1] 0.918 Clay) while spatial-Mondrian restores coverage to 0.955, 0.951, 0.956, 0.962, and 0.960 respectively, with paired-bootstrap CI excluding 0 for all five encoders (Fig. 4). The magnitude of repair tracks the debt: at this shared risk level the restoration set size is monotone in debt—1.62 (Prithvi) $>$ 1.58 (DOFA) $>$ 1.43 (SSL4EO-DINO) $>$ 1.39 (SSL4EO-MAE) $>$ 1.22 (Clay) (Fig. 5). The number of risk levels repaired is weakly monotone: Prithvi (heaviest debt) and DOFA (second-heaviest) are each repaired at $2/3$ levels ($\alpha = 0.05, 0.10$), while SSL4EO-DINO, SSL4EO-MAE, and Clay are each repaired at $1/3$ ($\alpha = 0.05$). At $\alpha = 0.20$ (nominal 0.80) none of the five is repaired because shifted accuracy (0.83–0.92, DOFA 0.85) already meets nominal—there is no under-coverage debt to repay, and the correction correctly does nothing (Fig. 4). DOFA was added to the same EuroSAT grid as a fifth encoder in a single-FM run whose spatial-Mondrian set sizes are $1.58/1.29/1.08$ at $\alpha = 0.05/0.10/0.20$; the run summary's aggregate KILL verdict is a multi-FM gate artifact (Clay and Prithvi were absent from that single-FM call), while the per-FM outcome is restoration at $\alpha = 0.05$ and 0.10 by the same paired-bootstrap CI-excludes-0 test applied to the four battery encoders (paired gap-difference CI low > 0), and no repair at $\alpha = 0.20$, where DOFA already over-covers (0.851). Correction note: this DOFA arm was re-run on 2026-07-13 after the initial 2026-07-12 run inadvertently probed DOFA's 45-way classification-head output instead of its 768-d pooled feature embedding; all DOFA numbers reported here are from the corrected embedding run, with the frozen buggy artifacts retained on disk for provenance. The four battery encoders are unaffected: their cached features were verified at their documented embedding widths (Prithvi and Clay 1024-d, SSL4EO-DINO 384-d, SSL4EO-MAE 768-d), and the bug's 45-dimensional signature appears in no other feature cache.

[1] SSL4EO-MAE's shifted split coverage is 0.8984 in this five-encoder run (shown as 0.898) and 0.8985 in the covariate- and label-shift baseline runs (Tables IV–V, shown as 0.899); the $<10^{-4}$ difference is independent calibration-split RNG on the same cached features, and split coverage equals shifted top-1 accuracy in every case (singleton sets).

Temperature scaling (37) proves insufficient: applying the source-fit temperature to the shifted logits lowers ECE to $0.058/0.041/0.054/0.032$ (Prithvi/SSL4EO-DINO/SSL4EO-MAE/Clay) from the raw $0.109/0.076/0.065/0.057$, but the residual still sits 2–11$\times$ above the in-distribution level ($0.026/0.014/0.005/0.013$), motivating the distribution-free conformal route; a focal-loss-based calibration alternative (38) is likewise post-hoc-parametric and is not explored here.

The headline is not a single-cut artifact: sweeping the source/target longitude cut over $P_{25}/P_{33}/P_{40}/P_{50}$ on the same cached embeddings (5 seeds each; Table VI) preserves the shift-ECE encoder ordering Prithvi $>$ SSL4EO-DINO $>$ SSL4EO-MAE $>$ Clay at every boundary (Prithvi $0.144/0.108/0.103/0.093$ vs. Clay $0.074/0.057/0.051/0.041$ at $P_{25}/P_{33}/P_{40}/P_{50}$), and the debt shrinks monotonically as the cut moves east and the target becomes less extreme (Prithvi split coverage $0.792 \to 0.859$). The ordering is over a small roster (five encoders at P_{33}, four across the sweep) and some adjacent gaps sit within seed noise—e.g. SSL4EO-DINO and SSL4EO-MAE differ by only $\approx$0.001 shift-ECE at P_{25}—so the extremes (Prithvi heaviest, Clay lightest) are robust while adjacent mid-rank ties should not be over-interpreted. At $\alpha = 0.05$, Mondrian restoration (paired-seed CI excluding 0) holds in 15 of 16 encoder$\times$boundary cells; the sole exception is Clay at P_{50}, where split coverage is already 0.936 (debt $\approx$0.014)—exactly the debt-targeted behaviour the conditional framing predicts. At $\alpha = 0.10$, Prithvi (heaviest debt) is restored at all four boundaries while the lighter-debt encoders mostly are not.

A label-access ablation shows that target calibration, not spatial binning, is the active ingredient: the non-spatial target-calibrated arm restores coverage to 0.946–0.952 at $\alpha = 0.05$ in every debt cell—statistically indistinguishable from spatial-Mondrian: the paired-seed "Mondrian minus target-global" gap-reduction CI excludes 0 in zero of 48 (FM$\times$boundary$\times\alpha$) cells, while target-global itself beats source-only split-conformal in the 22 cells where debt exists. Mondrian's set sizes are consistently larger (P_{33}, $\alpha = 0.05$: 1.62 vs. 1.56 Prithvi; 1.44 vs. 1.30 SSL4EO-DINO; 1.39 vs. 1.20 SSL4EO-MAE; 1.22 vs. 1.12 Clay). On this benchmark and 4×4 grid, then, the repair is bought by access to labeled target-region calibration data; per-spatial-bin conditioning adds no measurable marginal coverage benefit. We report this honestly and adjust the claim: the operative correction is *target-side conformal recalibration*, of which spatial-Mondrian is one (slightly costlier) instantiation here; finer spatial heterogeneity, where within-target variation is stronger, may still separate the two arms. Once spatial conditioning is inert, the remedy reduces to standard split-conformal calibrated on a small labeled target-region slice: we introduce no new conformal method, and spatial binning is reported here as an honest negative. The contributions of this paper are correspondingly the encoder-robustness diagnostic and the preregistered audit, not a new estimator.

Tables IV–VIII are a robustness battery, not five independent findings: each holds the frozen encoders and the coverage debt fixed and swaps exactly one variable—the correction method (unlabeled covariate-shift reweighting in Table IV; label-shift-adjusted reweighting in Table V), the source/target geographic cut (Table VI), or the benchmark itself (BigEarthNet's multi-label FNR arm in Table VII; the non-European So2Sat benchmark in Table VIII). In every table the single comparison worth making is the same one: at the shared $\alpha = 0.05$ row, read the source-only ("split") figure against the labeled target-side repair figure in that row—the gap between them is the debt each table is stress-testing, and it is the only number that would change the paper's claim if it closed.

A stronger unlabeled baseline still does not close the debt. Is the labeled target slice truly necessary, or would a principled unlabeled remedy suffice? As a post-hoc exploratory baseline on the cached EuroSAT features we add covariate-shift-weighted split conformal (17): density-ratio weights $w(x)$ from a logistic source-vs-target discriminator on the frozen embeddings, weighted-quantile calibration on the source scores, scored on the same shifted target (Table IV). Reweighting recovers part of the debt—at $\alpha = 0.05$ target coverage rises from source-split $0.833/0.880/0.899/0.918$ to $0.919/0.938/0.911/0.927$ (Prithvi/SSL4EO-DINO/SSL4EO-MAE/Clay)—but under-covers the nominal 0.95 for every encoder and never matches the labeled target-global arm (≈ 0.95). The mechanism is visible in the weights: the effective sample size is only 0.1–1.6% of the calibration set, i.e. the density ratios are so heavy-tailed that a handful of points dominate—a direct signature that this real E$\to$W shift is not the pure covariate shift under which weighted conformal is guaranteed (it also carries the class-prior shift documented in Methods). The honest reading reinforces the label-access finding: even a principled unlabeled correction is insufficient and unstable here, so a small labeled target slice remains the reliable remedy.

A label-shift-adjusted control rules out the class-prior confound: the P_{33} cut's severe class-prior shift (Table X) raises the possibility that the coverage debt above is a textbook label-shift effect (39) rather than a genuinely geographic one. We test this directly at the identical P_{33} split: we reweight the source calibration scores by $w(y) = \hat{\pi}_{\text{target}}(y)/\hat{\pi}_{\text{source}}(y)$ (Podkopaev–Ramdas label-shift-weighted split conformal), in an *oracle* variant (the true target-region label marginal, an upper bound on what prior-shift correction alone could repay) and a deployable variant (target priors from confusion-matrix Black-Box Shift Estimation (40), using unlabeled target predictions only). Neither restores coverage. For three of four encoders (Clay, SSL4EO-DINO, SSL4EO-MAE) both variants make coverage worse than doing nothing (e.g. Clay $0.918 \to 0.891$ oracle, 0.895 BBSE at $\alpha = 0.05$; Table V), and for Prithvi (heaviest debt) both recover only part of the gap ($0.833 \to 0.879/0.884$) while remaining far below both nominal and the labeled target-global arm (0.952). The overcorrection compounds at looser risk levels—at $\alpha = 0.20$ every encoder's label-shift-adjusted coverage sits 8.1–17.5 pp below nominal (Clay least at ≈ 8 pp, SSL4EO-DINO most at ≈ 17.5 pp)—and more than 15 pp below even the do-nothing source-split arm.[2] Because label-shift weighting is provably valid only when $P(x \mid y)$ is shift-invariant, this systematic failure rules out a pure class-prior-shift explanation and is consistent with a within-class conditional shift in the frozen encoders' features across regions—the geographic mechanism this paper posits, which target-side recalibration (which assumes nothing about $P(x \mid y)$) correctly repairs and label-shift reweighting cannot.

E. Replication on a second benchmark (BigEarthNet)

We replicate the phenomenon on a second real EO benchmark: GEO-Bench (26) `m-bigearthnet` (BigEarthNet-S2 (41); 7,669 usable tiles, real lat/lon, 36.96–67.80°N). We apply two complementary analyses.

A documented dominant-label single-label proxy (12 classes, ≥ 60 tiles each) lets the single-label LAC harness apply unchanged. All three frozen FMs replicate the pattern: ECE degrades ($0.36 \to 0.56$ Prithvi, $0.15 \to 0.42$ SSL4EO, $0.33 \to 0.59$ Clay) and spatial-Mondrian restores coverage to near-nominal at all three risk levels (Fig. 6). The proxy is a hard task (frozen-probe accuracy 0.50–0.61 in-distribution, dropping to 0.27–0.37 under shift), so restoration is bought at large prediction-set sizes ($\sim$6–10 of 12 classes)—a coarse, qualitative replication.

We extend the analysis with rigorous per-class binary conformal prediction on the 43-class multi-hot labels. For each of 39 well-populated labels (≥ 120 global positives; 27 active per seed after per-split filtering), we fit an independent binary probe (StandardScaler, L2-logistic, $C = 1$, 5 seeds) on source-region embeddings and calibrate per-label recall-coverage $\Pr(c \in \mathcal{C}(x) \mid y_c = 1)$ via source-only split-conformal and target-side spatial-Mondrian arms. The coverage-debt phenomenon replicates per-label: mean recall-coverage drops

[2] This control's *Split* and *Target-global* columns reproduce Table IV's values exactly; a covariate-shift-weighted reference column computed in the same run (not shown) differs slightly from Table IV because the two use different but both legitimate base-rate corrections for the density-ratio discriminator—a balanced-classifier posterior odds with no further correction here, versus an additional population-ratio ($n_{\text{src}}/n_{\text{tgt}}$) correction there—which affects only that reference column, not the label-shift arms.

TABLE IV

COVARIATE-SHIFT-WEIGHTED CONFORMAL (17) AS A POST-HOC EXPLORATORY UNLABELED BASELINE (EUROSAT E→W; 5-SEED MEANS; FULL $\alpha \in \{0.05, 0.10, 0.20\}$ SWEEP FROM THE SAME FROZEN RESULT RECORD, EXPANDING THE α=0.05 HEADLINE CELL). TARGET COVERAGE UNDER SOURCE-ONLY SPLIT, COVARIATE-SHIFT-WEIGHTED SPLIT (DENSITY-RATIO WEIGHTS FROM A SOURCE-VS-TARGET DISCRIMINATOR ON THE FROZEN EMBEDDINGS), AND THE LABELED TARGET-GLOBAL ARM, AGAINST NOMINAL $1-\alpha$. "ESS" IS THE WEIGHTS' EFFECTIVE SAMPLE SIZE AS A FRACTION OF THE SOURCE CALIBRATION SET; IT DOES NOT DEPEND ON α (THE WEIGHTS ARE FIXED BEFORE CALIBRATION), SO IT REPEATS PER ENCODER. ITS COLLAPSE SIGNALS THAT THE SHIFT IS NOT PURE COVARIATE SHIFT. WEIGHTED CONFORMAL HELPS BUT UNDER-COVERS NOMINAL AT EVERY α FOR EVERY ENCODER; ONLY THE LABELED TARGET-GLOBAL ARM RELIABLY REACHES IT.

α	Encoder	Split cov.	Weighted cov.	Target-global	ESS frac.
0.05	Prithvi	0.833	0.919	0.952	1.6%
0.05	SSL4EO-DINO	0.880	0.938	0.949	1.3%
0.05	SSL4EO-MAE	0.899	0.911	0.950	0.4%
0.05	Clay	0.918	0.927	0.951	0.1%
0.10	Prithvi	0.833	0.881	0.899	1.6%
0.10	SSL4EO-DINO	0.880	0.914	0.901	1.3%
0.10	SSL4EO-MAE	0.899	0.905	0.903	0.4%
0.10	Clay	0.918	0.923	0.918	0.1%
0.20	Prithvi	0.833	0.855	0.833	1.6%
0.20	SSL4EO-DINO	0.880	0.898	0.880	1.3%
0.20	SSL4EO-MAE	0.899	0.902	0.899	0.4%
0.20	Clay	0.918	0.921	0.918	0.1%

TABLE V

LABEL-SHIFT-ADJUSTED SPLIT CONFORMAL (39) AS A DECISIVE CONTROL FOR CLASS-PRIOR CONFOUNDING (EUROSAT E→W; 5-SEED MEANS; FULL $\alpha \in \{0.05, 0.10, 0.20\}$ SWEEP FROM THE SAME FROZEN RESULT RECORD, EXPANDING THE α=0.05 HEADLINE CELL). ORACLE USES THE TRUE TARGET-REGION LABEL MARGINAL; BBSE (40) ESTIMATES IT FROM UNLABELED TARGET PREDICTIONS ONLY. NEITHER RESTORES COVERAGE TO NOMINAL $1-\alpha$ AT ANY α; THREE OF FOUR ENCODERS ARE MADE WORSE THAN THE DO-NOTHING SPLIT ARM, AND THE DEFICIT WIDENS AT LOOSER α (MATCHING THE $\alpha = 0.20$ PP-DEFICIT NUMBERS QUOTED ABOVE).

α	Encoder	Split	LS-oracle	LS-BBSE	Target-global
0.05	Prithvi	0.833	0.879	0.884	0.952
0.05	SSL4EO-DINO	0.880	0.858	0.867	0.949
0.05	SSL4EO-MAE	0.899	0.856	0.865	0.950
0.05	Clay	0.918	0.891	0.895	0.951
0.10	Prithvi	0.833	0.793	0.798	0.899
0.10	SSL4EO-DINO	0.880	0.766	0.774	0.901
0.10	SSL4EO-MAE	0.898	0.793	0.803	0.903
0.10	Clay	0.918	0.825	0.832	0.918
0.20	Prithvi	0.833	0.675	0.682	0.833
0.20	SSL4EO-DINO	0.880	0.625	0.639	0.880
0.20	SSL4EO-MAE	0.898	0.688	0.704	0.898
0.20	Clay	0.918	0.711	0.719	0.918

from 0.92 in-distribution to 0.75 under shift ($\alpha = 0.10$; mean per-label debt 0.173), with spatial-Mondrian restoring to 0.91 (per-label paired CI excludes 0 at $\alpha \in \{0.10, 0.20\}$ for all three FMs; Fig. 7). Predicted-label set sizes remain large (20.9 of 27 active labels per tile at $\alpha = 0.10$), reflecting the geographic distributional mismatch and the recall requirement. Per-label conformal is more principled than the dominant-label proxy—it offers independent per-class recall guarantees—but both yield comparably large prediction sets under this hard pan-European shift.

This conformal-risk-control (CRC) arm, which gives a finite-sample FNR guarantee, is the substrate for the paper's primary decoupling result (Table I); here we give its full per-α calibration and repair numbers for all five frozen encoders. Per-label split-conformal offers per-class recall calibration but no finite-sample guarantee on the example-averaged false-negative rate (FNR). We therefore add a conformal-risk-control (CRC) arm (24) on the same frozen-feature caches, splits, 4×4 Mondrian grid, and per-label probes (per-example FNR loss, monotone in the threshold λ; exact weighted-quantile calibration; 5 seeds; percentile-bootstrap CIs, B=2000). The three-step story repeats under the marginal-FNR guarantee for every encoder (Table VII): (i) source-calibrated CRC meets the target FNR in-distribution (e.g. 0.105 at α=0.10, Prithvi); (ii) the same $\hat{\lambda}$ incurs FNR debt under the geographic shift (2–4× the nominal level, e.g. 0.206–0.277 at α=0.10); (iii) a Mondrian-CRC arm calibrated per spatial bin on the target-region slice restores marginal FNR to at-or-below nominal, at the cost of larger label sets (∼17–18 vs. ∼8–12 labels/tile at α=0.10). An honest efficiency note: CRC controls only the example-averaged FNR, so its in-distribution sets are smaller than per-label split-conformal's, and in-distribution FNR at α=0.05 fluctuates slightly above nominal (mean 0.052–0.056), consistent with the expectation-

TABLE VI
BOUNDARY-SENSITIVITY SWEEP, REAL-SHIFT TARGET COVERAGE "SPLIT / TARGET-GLOBAL / SPATIAL-MONDRIAN" (5-SEED MEANS; SPLIT = SOURCE-ONLY CALIBRATION; TARGET-GLOBAL = NON-SPATIAL, TARGET-CALIBRATED ABLATION ARM; FULL $\alpha \in \{0.05, 0.10, 0.20\}$ SWEEP FROM THE SAME FROZEN RESULT RECORD, EXPANDING THE α=0.05 HEADLINE ROW). † = MONDRIAN-VS-SPLIT PAIRED CI DOES NOT EXCLUDE 0 (NO SIGNIFICANT RESTORATION). AT α=0.05, RESTORATION HOLDS IN 15 OF 16 ENCODER×BOUNDARY CELLS (THE SOLE EXCEPTION, CLAY AT P_{50}, WHERE DEBT IS ≈0.014, IS MARKED); AT LOOSER α MOST CELLS NO LONGER NEED REPAIR BECAUSE SHIFTED ACCURACY ALREADY APPROACHES NOMINAL.

α	Encoder	P_{25}	P_{33} (headline)	P_{40}	P_{50}
0.05	Prithvi	0.792 / 0.946 / 0.952	0.833 / 0.952 / 0.955	0.843 / 0.950 / 0.955	0.859 / 0.949 / 0.952
0.05	SSL4EO-DINO	0.847 / 0.947 / 0.951	0.880 / 0.949 / 0.956	0.892 / 0.950 / 0.955	0.896 / 0.951 / 0.955
0.05	SSL4EO-MAE	0.865 / 0.948 / 0.957	0.898 / 0.950 / 0.962	0.910 / 0.946 / 0.962	0.917 / 0.952 / 0.967
0.05	Clay	0.897 / 0.950 / 0.955	0.918 / 0.951 / 0.960	0.926 / 0.947 / 0.960	0.936 / 0.950 / 0.963†
0.10	Prithvi	0.789 / 0.899 / 0.898	0.833 / 0.899 / 0.911	0.841 / 0.902 / 0.915	0.854 / 0.902 / 0.917
0.10	SSL4EO-DINO	0.847 / 0.897 / 0.906	0.880 / 0.901 / 0.926†	0.892 / 0.901 / 0.928†	0.896 / 0.902 / 0.930†
0.10	SSL4EO-MAE	0.865 / 0.895 / 0.920†	0.898 / 0.903 / 0.934†	0.910 / 0.910 / 0.938†	0.917 / 0.917 / 0.947†
0.10	Clay	0.897 / 0.900 / 0.920†	0.918 / 0.918 / 0.936†	0.926 / 0.926 / 0.935†	0.936 / 0.936 / 0.943†
0.20	Prithvi	0.789 / 0.802 / 0.832†	0.833 / 0.833 / 0.856†	0.841 / 0.841 / 0.867†	0.854 / 0.854 / 0.877†
0.20	SSL4EO-DINO	0.847 / 0.847 / 0.850†	0.880 / 0.880 / 0.888†	0.892 / 0.892 / 0.896†	0.896 / 0.896 / 0.902†
0.20	SSL4EO-MAE	0.865 / 0.865 / 0.880†	0.898 / 0.898 / 0.905†	0.910 / 0.910 / 0.917†	0.917 / 0.917 / 0.929†
0.20	Clay	0.897 / 0.897 / 0.897†	0.918 / 0.918 / 0.918†	0.926 / 0.926 / 0.926†	0.936 / 0.936 / 0.936†

level $\mathbb{E}[L] \leq \alpha$ guarantee—which holds in expectation over calibration draws, not per individual run, so a single run landing marginally above α is expected, not a violation. Like spatial-Mondrian conformal, Mondrian-CRC requires labeled target-region calibration data—a shift-aware remedy, not a free lunch.

F. External validation on a non-European benchmark (GEO-Bench m-so2sat)

Both benchmarks above are European Sentinel-2. To test whether the coverage-debt phenomenon and its target-side repair survive outside Europe, we repeat the harness on GEO-Bench (26) `m-so2sat` (So2Sat LCZ42 (42); 42 cities across multiple continents; 17 local-climate-zone classes), using GEO-Bench's own train/valid (source) vs. held-out test-city (target) partition as the shift. Because this release ships no per-tile latitude/longitude in our repo, there is no spatial-Mondrian arm here; we compare only source-only split-conformal against a non-spatial, target-calibrated arm (*target-global*), i.e. exactly the label-access ablation that Table VI isolated as the active ingredient on EuroSAT. All five frozen encoders now complete (Prithvi, Clay, SSL4EO ViT-S/16 MAE native-S2, SSL4EO-MAE ViT-B/16, DOFA); the earlier Clay v1.5 checkpoint-load failure (> 4 GB expected in cache) was resolved on the extraction host, so the encoder-ranking diagnostic is now available on this benchmark too (all five extracted on the hash-gated frozen substrate).

The debt-and-repair pattern replicates outside Europe: under the held-out-city shift, accuracy drops by ≈0.14–0.18 across the five encoders ($0.670 \rightarrow 0.522$ Prithvi up to $0.796 \rightarrow 0.637$ SSL4EO-MAE)—a genuine, non-trivial distribution shift. Source-only split-conformal under-covers at every risk level and every encoder ($\alpha = 0.05$: 0.877–0.904; $\alpha = 0.20$: 0.632–0.685), and the non-spatial target-calibrated arm restores coverage to near-nominal ($\alpha = 0.05$: 0.950–0.963), with the paired-seed restoration CI excluding 0 in every one of the fifteen debt cells (5 encoders $\times$ 3 risk levels; Table VIII). Restoration is bought at modest set sizes (∼2–6 of 17 classes), not vacuous full-label sets. This external label-access replication now spans all five encoders on a non-European, multi-continent benchmark: it confirms the operative part of the finding—*target-side conformal recalibration*, not spatial binning, repays the debt. It still lacks a spatial-Mondrian arm (no per-tile lat/lon), so it tests the label-access mechanism only; but with all five encoders now loadable it also carries the per-class conditional-coverage diagnostic (below). A data-provenance note applies: `m-so2sat` was obtained via GEO-Bench 1.0 (HuggingFace `recursix/geo-bench-1.0`) after the designed v0.9.1 Zenodo pull was unreachable, with a label map reconstructed from per-sample ground truth (21,964/21,964 samples, 0 failures); the v0.9.1→v1.0 content differences were not independently audited (see the data-provenance note in the code archive).

As a second, multiclass conditional-coverage dissociation, with all five encoders loadable, m-so2sat also lets us re-run the per-class conditional-coverage diagnostic of the EuroSAT conditional finding (§III) on a second, 17-class multiclass benchmark. At α=0.05 we measure, for the source-only split predictor, the worst individual LCZ class coverage per encoder (Table IX). It dissociates from both shifted accuracy and shift-ECE: $\mathrm{Spearman}(\text{worst-class cov}, \text{acc}) = -0.10$ (p=0.87, 5 pairwise inversions) and $\mathrm{Spearman}(\text{worst-class cov}, \mathrm{ECE}_{\mathrm{sh}}) = +0.60$ (p=0.29, 3 inversions); the class-coverage spread scrambles likewise ($\rho = -0.30$ against each, 6 inversions). The most accurate encoder here (SSL4EO-MAE, acc 0.637) is not the one that best protects its worst class, and the worst-covered class (0.689–0.733 across encoders, under marginal coverage 0.877–0.904) is under-served regardless of accuracy or calibration ranking. This corroborates the EuroSAT conditional-coverage finding with an independent multiclass benchmark—though on the unrepaired split sets here (no

TABLE VII

CRC ARM ON GEO-BENCH M-BIGEARTHNET (MARGINAL PER-EXAMPLE FNR; MEAN [95% BOOTSTRAP CI] POOLED OVER 5 SEEDS). SOURCE-CALIBRATED CRC MEETS NOMINAL IN-DISTRIBUTION, INCURS FNR DEBT UNDER THE REAL E→W SHIFT, AND TARGET-SIDE MONDRIAN-CRC RESTORES IT. SET SIZES ARE LABELS/TILE (OF ∼27 ACTIVE) UNDER SHIFT.

α	Encoder	CRC FNR in-dist	CRC FNR shift	Mondrian-CRC FNR shift	Sets CRC/Mond.
0.05	Prithvi	0.055 [0.051,0.058]	0.130 [0.125,0.135]	0.031 [0.029,0.034]	15.5 / 22.3
0.05	Clay	0.053 [0.049,0.056]	0.176 [0.171,0.182]	0.022 [0.020,0.024]	13.3 / 23.8
0.05	SSL4EO-DINO	0.053 [0.050,0.057]	0.191 [0.185,0.196]	0.032 [0.029,0.034]	10.6 / 22.3
0.05	SSL4EO-MAE	0.052 [0.048,0.055]	0.170 [0.164,0.175]	0.030 [0.028,0.033]	11.8 / 22.1
0.05	DOFA	0.056 [0.052,0.059]	0.169 [0.164,0.175]	0.021 [0.019,0.024]	13.0 / 24.1
0.10	Prithvi	0.105 [0.101,0.110]	0.206 [0.200,0.212]	0.083 [0.079,0.086]	11.9 / 18.2
0.10	Clay	0.108 [0.103,0.113]	0.277 [0.271,0.283]	0.082 [0.078,0.086]	9.7 / 18.2
0.10	SSL4EO-DINO	0.107 [0.102,0.112]	0.277 [0.271,0.283]	0.082 [0.078,0.086]	7.9 / 17.4
0.10	SSL4EO-MAE	0.108 [0.103,0.113]	0.262 [0.256,0.268]	0.084 [0.080,0.088]	8.6 / 17.1
0.10	DOFA	0.107 [0.103,0.112]	0.249 [0.243,0.256]	0.080 [0.076,0.084]	9.9 / 18.1
0.20	Prithvi	0.205 [0.199,0.211]	0.346 [0.339,0.353]	0.184 [0.179,0.190]	8.0 / 12.9
0.20	Clay	0.203 [0.197,0.209]	0.426 [0.419,0.433]	0.185 [0.179,0.190]	6.4 / 12.9
0.20	SSL4EO-DINO	0.206 [0.200,0.212]	0.401 [0.394,0.408]	0.191 [0.185,0.197]	5.3 / 11.1
0.20	SSL4EO-MAE	0.209 [0.202,0.215]	0.408 [0.401,0.415]	0.189 [0.184,0.195]	5.5 / 11.3
0.20	DOFA	0.203 [0.196,0.209]	0.384 [0.377,0.391]	0.187 [0.181,0.193]	6.7 / 12.4

TABLE VIII

NON-EUROPEAN EXTERNAL VALIDATION ON GEO-BENCH M-SO2SAT (17 LOCAL-CLIMATE-ZONE CLASSES, 42 MULTI-CONTINENT CITIES; 5-SEED MEANS). TARGET COVERAGE "SPLIT / TARGET-GLOBAL" UNDER THE HELD-OUT-CITY SHIFT (SPLIT = SOURCE-ONLY CALIBRATION; TARGET-GLOBAL = NON-SPATIAL, TARGET-CALIBRATED LABEL-ACCESS ARM; NO SPATIAL-MONDRIAN ARM, AS THIS RELEASE CARRIES NO PER-TILE LAT/LON IN OUR REPO). SET SIZES (OF 17 CLASSES) IN BRACKETS. ALL FIVE ENCODERS NOW LOAD (CLAY V1.5 CHECKPOINT-LOAD FAILURE RESOLVED).

α	Encoder	Split [set]	Target-global [set]	Acc. in→shift
0.05	Prithvi	0.904 [3.92]	0.956 [5.72]	0.670→0.522
0.05	Clay	0.891 [3.32]	0.954 [5.08]	0.713→0.569
0.05	SSL4EO ViT-S/16 MAE	0.880 [2.42]	0.962 [4.25]	0.781→0.601
0.05	SSL4EO-MAE ViT-B/16	0.882 [2.18]	0.950 [3.47]	0.796→0.637
0.05	DOFA	0.877 [2.49]	0.963 [5.35]	0.777→0.608
0.10	Prithvi	0.825 [2.71]	0.907 [4.02]	0.670→0.522
0.10	Clay	0.808 [2.27]	0.888 [3.24]	0.713→0.569
0.10	SSL4EO ViT-S/16 MAE	0.775 [1.67]	0.904 [2.75]	0.781→0.601
0.10	SSL4EO-MAE ViT-B/16	0.795 [1.57]	0.896 [2.38]	0.796→0.637
0.10	DOFA	0.781 [1.69]	0.918 [3.30]	0.777→0.608
0.20	Prithvi	0.685 [1.62]	0.819 [2.65]	0.670→0.522
0.20	Clay	0.672 [1.39]	0.795 [2.17]	0.713→0.569
0.20	SSL4EO ViT-S/16 MAE	0.632 [1.08]	0.809 [1.88]	0.781→0.601
0.20	SSL4EO-MAE ViT-B/16	0.654 [1.04]	0.793 [1.59]	0.796→0.637
0.20	DOFA	0.634 [1.07]	0.823 [1.99]	0.777→0.608

target-side repair or spatial arm on this release), and, like the EuroSAT case, it is post-hoc, exploratory, and small-n (n=5, all $p > 0.28$; §V).

IV. DISCUSSION

The through-line is that a conformal audit of a frozen GFM earns its keep exactly where accuracy and calibration go silent: on the multi-label FNR risk-control guarantee, every one of thirteen frozen encoders incurs a geographic FNR debt under shift (2.6–4.7× nominal) that only a labeled target slice repairs, and accuracy does not reliably say which encoder is most robust—the least accurate is the most robust, but at n=13 the rank correlation is a weak positive ($\rho = +0.32$; Table I)—and on per-class conditional coverage, where restoring the marginal guarantee still leaves one land-cover class covered at 0.850 under a 0.956 marginal (Table II), neither predictable from any scalar a practitioner would rank on. Symmetrically, we are explicit about where the audit does not add information: in the marginal single-label singleton regime the set-size axis is a deterministic restatement of accuracy, so EuroSAT at full data serves as a mapped negative control rather than a decoupling, and the preregistered efficiency-decoupling route is falsified. Within that boundary, the coverage-debt ranking remains a useful descriptive diagnostic. This secondary ranking is a debt-targeted phenomenon whose benefit is governed by the frozen encoder's calibration robustness, ordering Prithvi, SSL4EO-DINO, SSL4EO-MAE, and Clay consistently across risk levels. This ranking (and the headline coverage-restoration numbers throughout) is reported from an unadjusted logistic probe, i.e. not adjusting for the source/target class-prior mismatch the P_{33} cut induces (Methods; up to $6.54\times$ enrichment / $\sim 4.7\times$ depletion for individual EuroSAT classes). As a

TABLE IX
PER-CLASS CONDITIONAL COVERAGE ON GEO-BENCH M-SO2SAT (17 LCZ CLASSES; SOURCE-ONLY SPLIT-CONFORMAL, α=0.05; 5 SEEDS). THE WORST INDIVIDUAL-CLASS COVERAGE DISSOCIATES FROM SHIFTED ACCURACY AND SHIFT-ECE (Spearman $= -0.10/+0.60$; 5/3 PAIRWISE INVERSIONS), A SECOND MULTICLASS INSTANCE OF THE EUROSAT CONDITIONAL-COVERAGE COLLAPSE.

Encoder	Acc. shift	ECE shift	Marg. cov	Worst-class cov	Class spread
Prithvi	0.522	0.364	0.904	0.731	0.269
Clay	0.569	0.325	0.891	0.732	0.264
SSL4EO ViT-S/16 MAE	0.601	0.114	0.880	0.689	0.302
SSL4EO-MAE ViT-B/16	0.637	0.173	0.882	0.702	0.269
DOFA	0.608	0.214	0.877	0.733	0.253

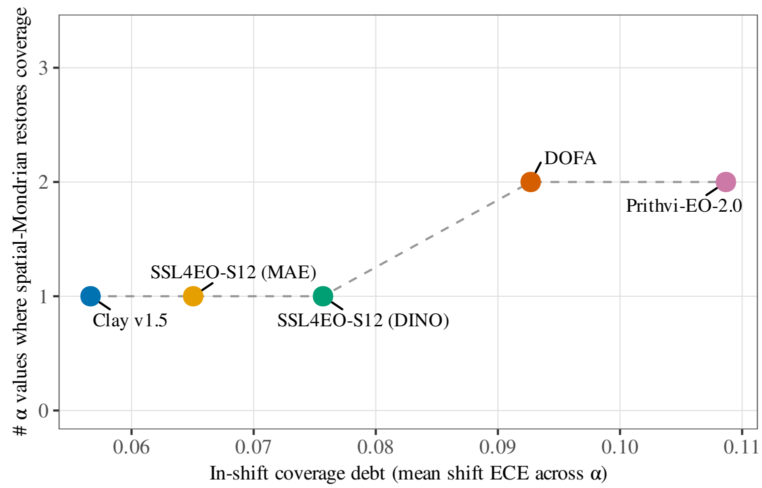

Fig. 3. Encoder-dependent coverage debt (mean shift ECE) vs. the number of risk levels spatial-Mondrian conformal restores, for all five frozen encoders. Debt orders Prithvi > DOFA > SSL4EO-DINO > SSL4EO-MAE > Clay; repair count is weakly monotone (Prithvi = DOFA 2; SSL4EO-DINO = SSL4EO-MAE = Clay 1), and the repair magnitude (Fig. 5) is monotone in debt. The five frozen encoders comprise the four-encoder battery plus the corrected DOFA fifth-encoder block.

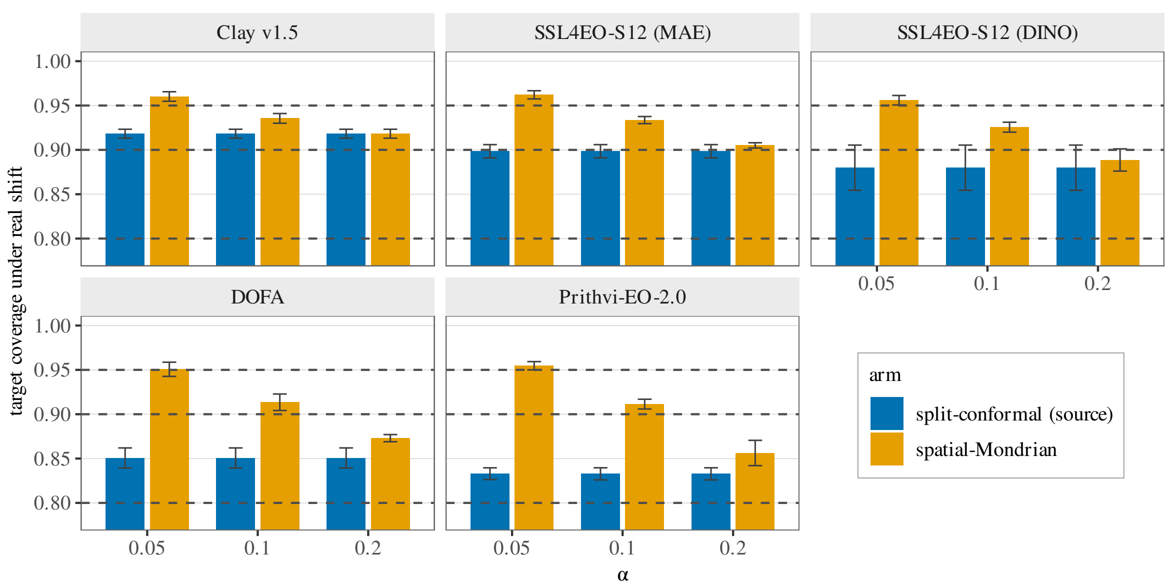

Fig. 4. Target coverage under real shift: source split-conformal (under-covers) vs. spatial-Mondrian, per encoder and α (dashed = nominal $1-\alpha$; error bars seed-level t-CIs, n=5), for all five frozen encoders (roster as in Fig. 3). Restoration occurs where debt exists; DOFA's shifted split coverage is 0.851 at every α and spatial-Mondrian lifts it to 0.951/0.913/0.873.

control, we re-fit every probe with class-balanced sample weighting on the identical cached embeddings (Table XI): the encoder ranking by shift-ECE, the per-encoder count of risk levels repaired, and the analogous per-label-balanced

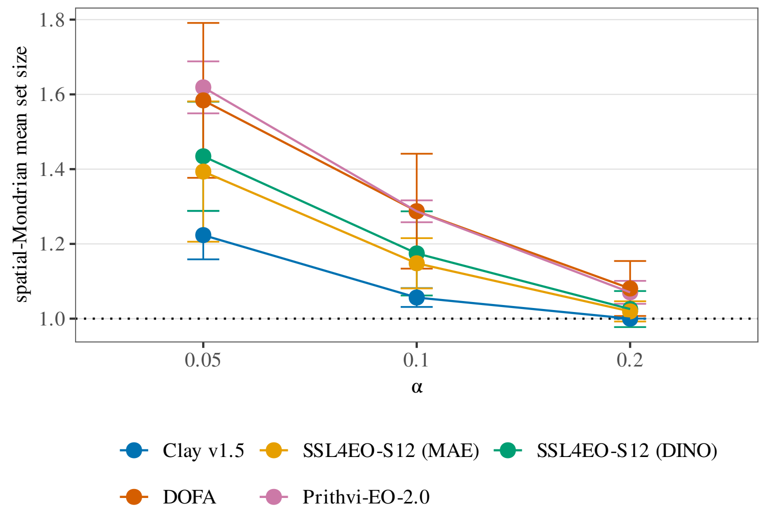

Fig. 5. Restoration is paid in prediction-set size: spatial-Mondrian set size exceeds the split-conformal singleton baseline (dotted line at set size 1) only where it repays coverage debt, across all five frozen encoders (roster as in Fig. 3; DOFA's set size $1.58/1.29/1.08$ at $\alpha = 0.05/0.10/0.20$). Error bars are seed-level t-CIs (n=5).

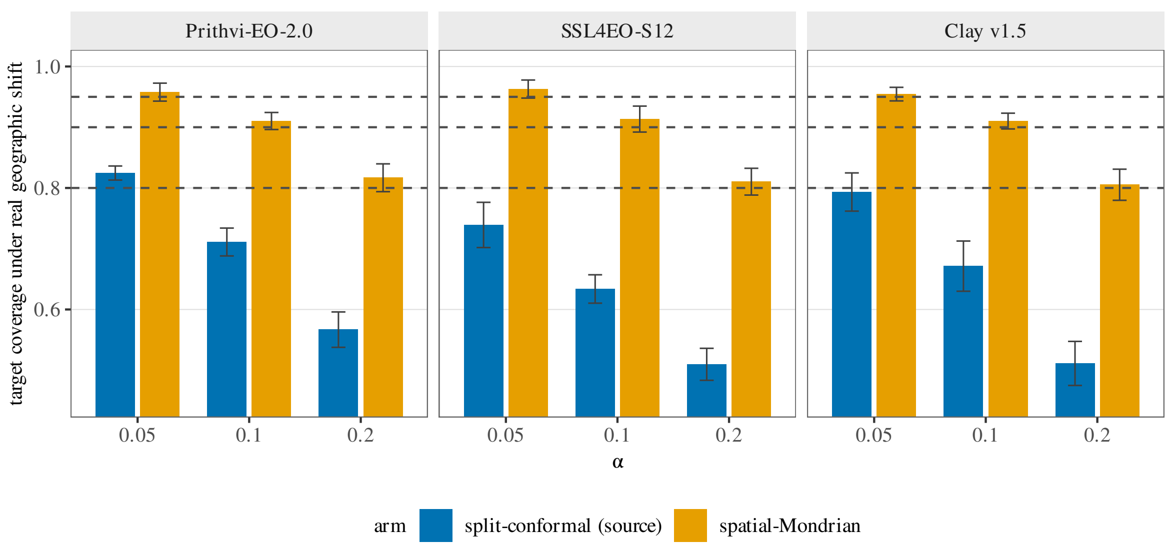

Fig. 6. Dominant-label proxy on GEO-Bench `m-bigearthnet` (single-label reduction; 12 classes): coverage debt and spatial-Mondrian restoration replicate across all three frozen FMs (dashed = nominal $1 - \alpha$; error bars seed-level t-CIs, n=5). A coarse, qualitative replication; the per-label analysis of Fig. 7 is primary.

check on the BigEarthNet per-label conformal analysis are all unchanged to within the seed noise, so the class-prior mismatch does not appear to be driving the ranking—but the unadjusted numbers remain the ones reported as the headline throughout, and the balanced control was run only at the single P_{33} boundary (the boundary sweep of Table VI uses the unadjusted probe). Notably, the ViT-B backbone (SSL4EO-MAE, 768-dim) incurs less debt than the ViT-S backbone (SSL4EO-DINO, 384-dim); however, these two checkpoints differ simultaneously in architecture size, pretraining objective (MAE vs. DINO), and specific checkpoint (Methods), so this pairwise comparison is consistent with, but does not isolate, a backbone-capacity mechanism for geographic calibration robustness (this pairwise ordering is also unchanged under the balanced-probe control). We deliberately avoid a "universal fix" claim: where an encoder is robust enough that shifted accuracy meets the nominal level, there is no debt and no repair.

A. *Class-prior shift and a balanced-probe control*

The encoder ordering by shift-ECE (Prithvi $>$ SSL4EO-DINO $>$ SSL4EO-MAE $>$ Clay) and the number of risk levels each encoder's spatial-Mondrian arm repairs are identical under the balanced control. The analogous per-label-balanced refit of the BigEarthNet per-label conformal probes (same cached embeddings, $\alpha = 0.10$) likewise leaves mean per-label debt and restoration essentially unchanged (e.g. Prithvi debt

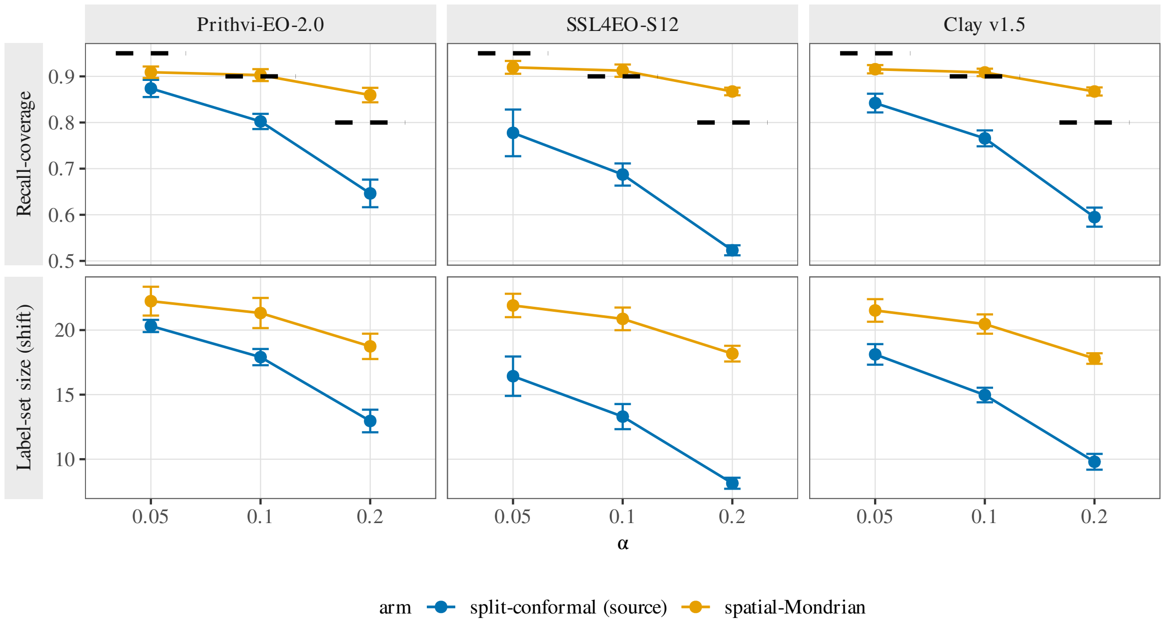

Fig. 7. Per-label multi-label conformal on GEO-Bench (26) `m-bigearthnet` (BigEarthNet-S2; 27 active labels; real geographic E→W shift; 5 seeds). Top: mean per-label recall-coverage $\Pr(c \in \mathcal{C}(x) \mid y_c = 1)$ for source split-conformal and target-side spatial-Mondrian per FM and α (dash $= 1-\alpha$). Bottom: mean predicted label-set size per tile. Mondrian restores per-label coverage at $\alpha \in \{0.10, 0.20\}$ (paired CI across labels excludes 0) for all three FMs, confirming coverage debt in a rigorous multi-label setting.

TABLE X

EUROSAT-MS PER-CLASS FREQUENCY AT THE P_{33} LONGITUDE CUT (SOURCE = E/CENTRAL EUROPE, n=18,090; TARGET = W/IBERIAN-ATLANTIC EUROPE, n=8,910). CLASSES NOT LISTED SHIFT BY <2.4×.

Class	Source %	Target %	Target/Source ratio
5 (Pasture)	2.62	17.13	6.54× (enriched)
2 (HerbaceousVegetation)	7.68	18.07	2.35× (enriched)
9 (SeaLake)	14.10	5.04	0.36× (depleted)
8 (River)	11.60	4.51	0.39× (depleted)
1 (Forest)	15.02	3.16	0.21× (∼4.7× depleted)

TABLE XI

CLASS-PRIOR-SHIFT CONTROL: CLASS-BALANCED SAMPLE WEIGHTING VS. THE PAPER'S UNADJUSTED LOGISTIC PROBE, RE-FIT ON THE IDENTICAL CACHED FROZEN EMBEDDINGS (NO GPU RE-EXTRACTION), 5 SEEDS, REAL E→W SHIFT. TOP PANEL (EUROSAT, $\alpha = 0.05$): ECE-SHIFT RANKING AND PER-ENCODER REPAIR COUNT (OF 3 α'S) ARE UNCHANGED UNDER THE BALANCED CONTROL. BOTTOM PANEL (BIGEARTHNET-S2, $\alpha = 0.10$, PER-LABEL CONFORMAL): THE ANALOGOUS BALANCED-PROBE REFIT OF THE PER-LABEL HARNESS (PREVIOUSLY REPORTED ONLY IN PROSE) LEAVES MEAN PER-LABEL DEBT AND MONDRIAN RESTORATION ESSENTIALLY UNCHANGED FOR ALL THREE AVAILABLE FMS (SSL4EO-MAE WAS NOT RUN IN THE PER-LABEL BIGEARTHNET HARNESS).

	Unadjusted (paper headline)			Balanced control		
Encoder	Acc. shift	ECE shift	#α repaired	Acc. shift	ECE shift	#α repaired
Prithvi	0.833	0.1085	2/3	0.835	0.1111	2/3
SSL4EO-DINO	0.880	0.0755	1/3	0.878	0.0798	1/3
SSL4EO-MAE	0.898	0.0650	1/3	0.895	0.0708	1/3
Clay	0.918	0.0568	1/3	0.917	0.0584	1/3

	Unadjusted (paper headline)		Balanced control	
Encoder	Mean per-label debt	Mondrian restores	Mean per-label debt	Mondrian restores
Prithvi	0.117	Yes	0.113	Yes
SSL4EO-DINO	0.239	Yes	0.236	Yes
Clay	0.163	Yes	0.161	Yes

0.117 → 0.113, SSL4EO 0.239 → 0.236, Clay 0.163 → 0.161; restoration CI still excludes 0 for all three FMs). Full per-seed numbers, the per-class frequency table for all 10 classes, and the BigEarthNet balanced-control results are included in the code archive.

B. A deployment report card

An illustration of the audit's operational use (post-hoc; not a preregistered claim). Concretely, suppose an agency must monitor multi-label land cover in an unseen region with a frozen off-the-shelf GFM at risk level α=0.05, and picks its encoder the way a practitioner would—on accuracy. The median-accuracy encoder (Clay v1.5; in-distribution mAP 0.502, rank 3 of 5) is a deliberately unremarkable choice. Table XII is the reliability report card the audit returns for it, assembled entirely from the frozen result records (no new computation). Deployed uncalibrated, its example-averaged FNR guarantee is missed by $\approx 3.5\times$ (0.176 vs. the promised 0.05); a 30% labeled target-region calibration slice with target-side Mondrian CRC restores it to 0.022, at the cost of larger prediction sets ($13.3 \rightarrow 23.8$ labels of 27). A second, non-European region (GEO-Bench `m-so2sat`, LCZ-17) shows the complementary caution: a healthy marginal coverage can still hide a land-cover class covered at only 0.732. The recommended operating point is therefore to deploy this encoder only with a labeled target-region recalibration slice, and to monitor per-class coverage, not merely the marginal.

V. Limitations

This section is a single continuous preregistration-disclosure narrative, not a conventional caveats list, and it is deliberately long because it is by design as load-bearing as the results themselves. It discloses, in order: (i) the paper's one preregistered outcome and how the encoder-dependent claim reported in the main text reframes it (the Stage-2 *universal-restoration* gate); (ii) the power limits of every post-hoc decoupling analysis added after that gate, including the roster-expansion result that weakened rather than strengthened, and the small-n conditional-coverage findings; and (iii) the audit's scope boundaries—single-sensor, single-continent until the So2Sat extension, and encoder-roster size—together with the negative-control battery run to rule out confounds. A first-time reader should treat what follows as the paper's own accounting of what its evidence does and does not license, not as an appendix of afterthoughts.

As a preregistration disclosure and exploratory reframe: the Stage-2 analysis was preregistered with a *universal-restoration* success criterion: spatial-Mondrian restoration at ≥ 2 of 3 risk levels for both Prithvi and Clay (H3–H4). The machine gate for that criterion returned KILL, because Clay was restored at only 1 of 3 levels (Prithvi passed with 2/3). The conditional, encoder-dependent claim reported throughout this paper is therefore an exploratory reframe of that preregistered outcome—the debt-targeted framing was adopted after observing that repair occurs exactly where debt exists—and should be read as hypothesis-generating rather than confirmatory; the original gate verdict is preserved verbatim in the result JSON. In confirmatory terms, the single preregistered result this paper delivers is the falsification of the universal-restoration hypothesis (the KILL); the encoder-dependent account built on top of it is hypothesis-generating. A distinct firm positive stands beside that negative: the within-encoder FNR-debt-and-repair pattern is not itself a preregistered population claim, but a distribution-free sign test computed post-hoc over the thirteen frozen per-encoder estimates confirms it at $p \approx 2.4 \times 10^{-4}$ in both the debt and the repair direction (§III). In scope terms, this paper introduces no new estimator: we reuse split and spatial-Mondrian conformal prediction (22, 33) and Angelopoulos–Bates conformal risk control (24) throughout, and the operative axis is access to a small labeled target-region calibration slice, not a new method—the contribution is the applied diagnostic and its validated repair, not conformal machinery.

The re-centered decoupling axes are post-hoc and (for the conditional analyses) small-n: the three analyses—the multi-label FNR decoupling, the per-class conditional-coverage collapse, and the singleton-degeneracy boundary—were all run after the preregistered gate, on the same frozen-feature caches, and are exploratory. Two power caveats bound them. First, the cross-encoder FNR comparison is the one claim the roster expansion actively weakened, and we report the weakened result. Over the five-encoder battery the shifted FNR rank-correlated with mAP at $+0.90$ (asymptotic p=0.037, though its *exact* permutation p was already 0.083); expanding to the full n=13 roster regresses the statistic to $+0.32$ (permutation p=0.28), with 48/78 (62%) pairwise inversions. The strong five-encoder inversion therefore does *not* generalize—we keep only the weaker, directional claim that accuracy is not a reliable guide to FNR robustness: the correlation stays positive (it never reverses to "better encoder $\rightarrow$ better reliability") and the extreme case survives (Prithvi, least accurate, is the most FNR-robust; the near-calibration-matched Prithvi/Clay pair still dissociates, ECE 0.004 apart yet 0.046 apart in FNR and in the wrong direction), but at n=13 this is directional evidence, not a proven inversion. The one companion correlation that survives is the FNR-vs-set-size tradeoff ($\rho = -0.66$, p=0.017 at α=0.10): robustness is bought with larger sets. Second, the conditional-coverage dissociation is over n=5 encoders with all $p > 0.28$ and a rank correlation that flips sign across α; it rests on the SSL4EO-DINO datum (one class at 0.850 under a 0.956 marginal, the same class in all five seeds, per-seed range 0.826–0.871, $2\times$ the next-worst) plus 6/10 inversions, so the honest claim is "no detectable/stable dependence," not proven independence. Both are strong concrete decouplings that a practitioner cannot anticipate from accuracy or ECE, and both would benefit from a larger encoder roster—a scoping choice (frozen-feature extraction only), not a capability gap. The singleton-degeneracy boundary, by contrast, is a deterministic property we simply map, not a correlation.

The headline shifts are cross-region (single-sensor Sentinel-2), not cross-sensor. BigEarthNet confirms the phenomenon under both the dominant-label proxy and rigorous per-label conformal; the two analyses agree qualitatively and the per-label treatment is now primary. A preliminary cross-sensor probe (Sentinel-2 calibration transferred to a native Sentinel-1 encoder on paired m-so2sat tiles) is withheld: its in-distribution control is anomalous— S2 accuracy collapses to chance (0.06) even in-distribution, versus 0.513 for the same encoder on the identical subset in a companion run, indicating an undiagnosed run-specific failure—a debug pass

TABLE XII
DEPLOYMENT RELIABILITY REPORT CARD (ILLUSTRATION; POST-HOC) FOR A MEDIAN-ACCURACY FROZEN ENCODER (CLAY V1.5) MONITORING LAND COVER IN AN UNSEEN REGION AT α=0.05. MULTI-LABEL FNR ROWS: GEO-BENCH `M-BIGEARTHNET`, $E\to W$ SHIFT; WORST-CLASS ROW: GEO-BENCH `M-SO2SAT` LCZ-17 (SPLIT ARM). BRACKETS ARE SEED-LEVEL t-CIS (n=5). ALL VALUES TRACE TO THE FROZEN RESULT RECORDS DESCRIBED IN THE CODE ARCHIVE.

Report-card question (α=0.05, encoder Clay)	Value
Multi-label FNR if deployed uncalibrated	0.175 $[0.143, 0.208]$
(relative to the 0.05 target)	$\approx$3.5$\times$ over
FNR after a 30% target slice (Mondrian CRC)	0.022 $[0.015, 0.029]$
Efficiency cost (mean set size, of 27 labels)	$13.3 \to 23.8$
Worst class, 2nd region (`m-so2sat` LCZ)	0.732 $[0.692, 0.772]$
(under a marginal coverage of)	0.891

exonerated the resampling code but could not re-extract the features to pin the exact trigger—rather than a sensor effect. A contrastively-aligned S1/S2 encoder is the right vehicle for the cross-sensor question and is left to future work; we make no cross-sensor claim here. Disentangling backbone size from pretraining objective would require a ViT-S-MAE and a ViT-B-DINO checkpoint, a 2×2 cell only partially available (the public SSL4EO-S12 release contains no ViT-B/16+DINO Sentinel-2 checkpoint); we leave it to future work. On encoder diversity, a fifth distinct architectural family (DOFA) has now been added to the headline EuroSAT grid (§III), where it carries the second-heaviest coverage debt of the five—between Prithvi and SSL4EO-DINO—and is restored at $2/3$ risk levels, consistent with the encoder ordering reported here. (Its GEO-Bench `m-brick-kiln` cell used the same pre-correction feature path and is withheld pending re-computation; that task's v1.0 data was not available on the correction host.) The encoder roster is illustrative of the diagnostic, not a leaderboard: adding a further 2025-generation family such as Galileo/TerraFM needs only frozen-feature extraction—no fine-tuning—and is a scoping choice rather than a capability gap. On external geographic generalization, the single-continent limitation is now partly addressed: the non-European `m-so2sat` validation above (42 cities, multiple continents) reproduces the debt-and-repair pattern across all $5/5$ encoders and $3/3$ risk levels (Table VIII; the earlier Clay checkpoint-load block is resolved), and now supplies a second multi-class conditional-coverage dissociation (Table IX), though it remains a label-access ablation without a spatial-Mondrian arm. EuroSAT-MS and the BigEarthNet-S2 extension (36.96–67.80°N) remain European Sentinel-2, so continent-scale generalization of the spatial-Mondrian arm specifically is still future work. The EuroSAT-MS headline numbers were fixed at a single longitude cut (P_{33}) before any alternative was tried; the boundary-sensitivity sweep of Table VI is post-hoc (run 2026-07-02 on the cached embeddings) and confirms the encoder ordering and debt-targeted restoration at $P_{25}/P_{40}/P_{50}$, so the n=1-boundary caveat is now resolved empirically, though the headline numbers themselves remain those of the pre-committed P_{33} cut. Finally, the label-access ablation shows that on this benchmark the non-spatial target-calibrated baseline matches spatial-Mondrian, so our results do not demonstrate a benefit of spatial conditioning per se. Two adjusted-conformal baselines a hostile reading would demand have now been run as negative controls: covariate-shift-weighted conformal (Table IV) under-covers for every encoder, and label-shift-adjusted conformal (oracle and BBSE, Table V) fails to restore coverage and worsens it for three of four encoders—ruling out the class-prior-shift confound the P_{33} cut could otherwise raise. Class-conditional Mondrian conformal remains unexecuted.

DATA AND CODE AVAILABILITY

EuroSAT-MS (public, Sentinel-2); frozen weights Prithvi-EO-2.0-300M, Clay v1.5, SSL4EO-S12 DINO (torchgeo), SSL4EO-S12 MAE (`wangyi111/SSL4EO-S12`, HuggingFace), all public/no-auth. The code and frozen result records used to produce every result and figure in this manuscript are available at GitHub (`https://github.com/PeterPonyu/geospatial-fm-reliability-research`) and archived at Zenodo, DOI `10.5281/zenodo.21130299` (this DOI is currently reserved on a draft deposition and becomes active when the record is published); reproduction instructions and a fast regression gate are provided in the code archive. Per IEEE's author-reproducibility guidelines, code archiving with a persistent identifier via GitHub plus a DOI-issuing repository (Zenodo) is sufficient for JSTARS; IEEE's optional Code Ocean partnership was not used for this submission.

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